

FUME EXTRACTOR

CLEAN AIR FOR MAKERS

A compact, fully 3D-printed fume extractor designed for soldering and electronics work. Built using an **Arduino Nano**, the system allows adjustable fan speed through a **potentiometer** while a **servo motor** changes the extraction angle for better airflow control.

Fully
3D Printed:



FEATURES

-  Fully 3D Printed Body
-  Arduino Nano Controlled
-  Adjustable Fan Speed
-  Servo-Based Angle Control
-  Compact Desktop Design
-  Built for Electronics & Soldering

COMPONENTS

-  Arduino Nano
-  Servo Motor
-  Potentiometer
-  High-Speed Fan
-  Custom 3D Printed Enclosure
-  Toggle Controls

SCAN ME



HOW IT WORKS >>>>>>>>



The **potentiometer** acts as the speed controller for the extraction fan, allowing airflow adjustment depending on the amount of smoke or fumes produced.



A **servo motor** mounted on the side rotates the extractor head, changing the angle of airflow direction for improved efficiency and positioning flexibility.



The entire system runs through an **Arduino Nano**, coordinating both the fan control and servo movement.

BUILT BY

VIDIT BHATI

16-YEAR-OLD MAKER
FROM INDIA



DESIGN LANGUAGE

Inspired by industrial workshop equipment and modern cyber-tech aesthetics, the design focuses on functionality, portability, and clean airflow management for makerspaces.



FUNCTIONAL



COMPACT



EFFICIENT



MAKER-FOCUSED